LESSON 01 / 07

Why one number controls the whole subject

A **quadratic equation** is $ax^2 + bx + c = 0$ with $a \neq 0$ (if $a = 0$ it is linear, and every formula below collapses). Coefficients are real unless stated otherwise. The equation asks: where does the parabola $y = ax^2 + bx + c$ meet the x-axis? Everything — nature of roots, sign of the expression, location of roots — is bookkeeping built on one quantity.

SHORT DERIVATION

Completing the square forces the formula and the discriminant

Divide by a and complete the square:

$$x^2 + \frac{b}{a}x = -\frac{c}{a} \quad \Rightarrow \quad \left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

The left side is a square, so a real solution exists iff the right side is ≥ 0, i.e. iff $D = b^2 - 4ac \geq 0$. Taking roots gives the **quadratic formula**

$$x = \frac{-b \pm \sqrt{D}}{2a}, \qquad D = b^2 - 4ac$$

D is called the **discriminant** because it discriminates between the cases. Note it is $b^2 - 4ac$, never $b^2 + 4ac$.

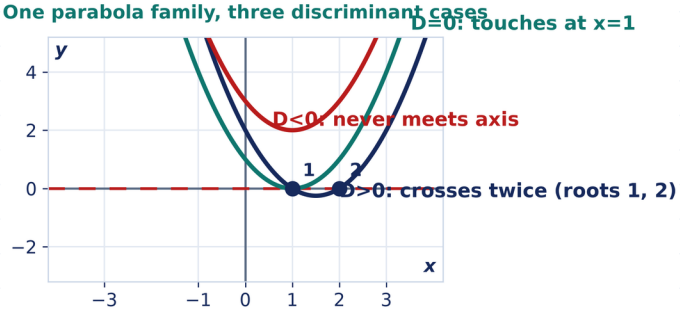
KEY RESULT

The three cases — memorise with the picture

- $D > 0$: two **distinct real** roots; the parabola crosses the axis twice.
- $D = 0$: one **repeated** real root $x = -\frac{b}{2a}$; the parabola touches the axis at its vertex.
- $D < 0$: **no real roots**, but two complex roots, always a conjugate pair $\alpha \pm i\beta$ (real coefficients force this: conjugating the equation changes nothing, so $\bar{\alpha}$ is a root whenever α is).

Saying "$D < 0$ means no roots" is wrong — it means no real roots.

One parabola family, three discriminant cases



WORKED EXAMPLE

Worked — reading the case off D

$x^2 - 5x + 6 = 0$: $D = 25 - 24 = 1 > 0$, so $x = \frac{5 \pm 1}{2}$, giving $x = 2, 3$.

$2x^2 - 3x + 5 = 0$: $D = 9 - 40 = -31 < 0$, so $x = \frac{3 \pm i\sqrt{31}}{4}$ — non-real conjugates.

Equal roots: $x^2 + kx + 9 = 0$ needs $D = k^2 - 36 = 0$, so $k = \pm 6$ (both signs — a classic one-mark loss).

Rational, irrational, and conjugate root pairs

KEY RESULT

What the coefficients force the roots to look like

- Rational coefficients and D a perfect square (of a rational) $\Leftrightarrow$ both roots **rational**.
- $D > 0$ but not a perfect square: roots are irrational **conjugates** $p \pm \sqrt{q}$ with p, q rational.
- Real coefficients, $D < 0$: complex conjugates $\alpha \pm i\beta$.
- One root known of conjugate type forces the other: if $2 + \sqrt{3}$ is a root of a rational-coefficient quadratic, $2 - \sqrt{3}$ is the other root. Product $= 4 - 3 = 1$, sum $= 4$, so the equation is $x^2 - 4x + 1 = 0$.

COMMON TRAP

Common traps — foundations

- $D = b^2 - 4ac$: the minus sign, and the 4.
- "Equal roots" means $D = 0$ **and** the equation is genuinely quadratic ($a \neq 0$). If a contains the parameter, check the degenerate case separately.
- If coefficients depend on a parameter, $D \geq 0$ alone does not guarantee two real roots when the equation can degenerate to linear.
- A repeated root is still a root of multiplicity two — it counts twice in Vieta's relations.

ASSUMPTIONS & CONVENTIONS

Standing conventions for this chapter

$a, b, c \in \mathbb{R}$, $a \neq 0$; roots denoted α, β; $S = \alpha + \beta$, $P = \alpha\beta$; "roots" means complex roots counted with multiplicity unless "real roots" is said. All root-location conditions in sheet 04 assume real coefficients.

LESSON 02 / 07

The two relations that do all the work

SHORT DERIVATION

Vieta from factorising

If α, β are the roots, then $ax^2 + bx + c = a(x - \alpha)(x - \beta) = a(x^2 - (\alpha +$. Comparing coefficients:

$$S = \alpha + \beta = -\frac{b}{a}, \qquad P = \alpha\beta = \frac{c}{a}$$

Watch the signs: sum carries the minus, product does not. The derivation also shows why Vieta survives in $\mathbb{C}$: factorisation needs no reality.

KEY RESULT

The symmetric-function toolkit (all from S and P)

$$\alpha^2 + \beta^2 = S^2 - 2P, \qquad \alpha^3 + \beta^3 = S^3 - 3PS$$

$$\alpha^2\beta + \alpha\beta^2 = PS, \qquad \frac{1}{\alpha} + \frac{1}{\beta} = \frac{S}{P} \quad (P \neq 0)$$

$$(\alpha - \beta)^2 = S^2 - 4P = \frac{D}{a^2}, \qquad |\alpha - \beta| = \frac{\sqrt{D}}{|a|} \ (D \geq 0)$$

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{S^2 - 2P}{P}, \qquad \alpha^2 - \beta^2 = (\alpha - \beta)S$$

Deep reason: swapping $\alpha \leftrightarrow \beta$ changes no coefficient, so any symmetric expression must be expressible through S, P alone. $\alpha - \beta$ is **not** symmetric — that is exactly why the discriminant has to be brought in for it.

WORKED EXAMPLE

Worked — everything from S and P, roots never found

For $x^2 - 5x + 6 = 0$: $S = 5$, $P = 6$. Then $\alpha^2 + \beta^2 = 25 - 12 = 13$; $\alpha^3 + \beta^3 = 125 - 90 = 35$; $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{5}{6}$; $(\alpha - \beta)^2 = 25 - 24 = 1$; $\alpha^2\beta + \alpha\beta^2 = 30$; $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{13}{6}$. (Roots are $2, 3$ — check: $4 + 9 = 13$, $8 + 27 = 35$ ✓.)

For $3x^2 - 7x + 2 = 0$: $S = \frac{7}{3}$, $P = \frac{2}{3}$ directly — solving $(x = 2, \frac{1}{3})$ is wasted effort.

COMMON TRAP

Common traps — Vieta

- $(\alpha - \beta)^2 = S^2 - 4P$, not $S^2 - 2P$ (that is $\alpha^2 + \beta^2$). The cross-term doubles.
- $|\alpha - \beta| = \sqrt{S^2 - 4P}$ needs $D \geq 0$ to be a real distance; over $\mathbb{C}$ only $(\alpha - \beta)^2$ is meaningful.
- Reciprocal identities need $P \neq 0$, i.e. $c \neq 0$: a zero root has no reciprocal.
- $\alpha - \beta$ has a sign ambiguity; problems asking for it usually want $|\alpha - \beta|$ or $(\alpha - \beta)^2$.

Forming an equation from its roots

KEY RESULT

The reverse direction

Roots $r_1, r_2 \Rightarrow$ equation $x^2 - (r_1 + r_2)x + r_1 r_2 = 0$ (monic form; multiply out to clear fractions). This is the whole of "form the equation whose roots are ...": compute the **new sum** and **new product**, never the roots themselves.

Roots α^2, β^2 for $x^2 - 5x + 3 = 0$ ($S = 5, P = 3$): new sum $= S^2 - 2P = 19$, new product $= P^2 = 9$, so $x^2 - 19x + 9 = 0$. Note product P^2, not P — checklist item.

STANDARD APPLICATION

Where this is used everywhere else

Condition problems ("roots in ratio $m:n$", "one root is the square of the other", "roots differ by k") are all Vieta plus one extra equation. E.g. roots differing by 1 for $x^2 - 5x + 6$: $(\alpha - \beta)^2 = S^2 - 4P = 1$ gives the difference without solving. Ratio $m:n$: set roots mt, nt, then $S = (m+n)t$ and $P = mnt^2$, so $P(m+n)^2 = mnS^2$ — eliminate t immediately.

LESSON 03 / 07

The expression as a function

SHORT DERIVATION

Completed-square form reads off everything

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 - \frac{D}{4a}$$

The square term is ≥ 0, so the extremum sits at the **vertex** $x_V = -\frac{b}{2a}$ and equals $-\frac{D}{4a} = \frac{4ac - b^2}{4a}$: a **minimum** if $a > 0$, a **maximum** if $a < 0$. The axis of symmetry is $x = -\frac{b}{2a}$.

WORKED EXAMPLE

Worked — vertex without calculus

$x^2 - 6x + 11 = (x-3)^2 + 2$: minimum 2 at $x = 3$. $-x^2 + 4x - 1 = -(x-2)^2 + 3$: maximum 3 at $x = 2$. For $y = 2x^2 - 8x + 5$: $x_V = \frac{8}{4} = 2$, $y_V = \frac{40 - 64}{8} = -3$; vertex $(2, -3)$, axis $x = 2$.

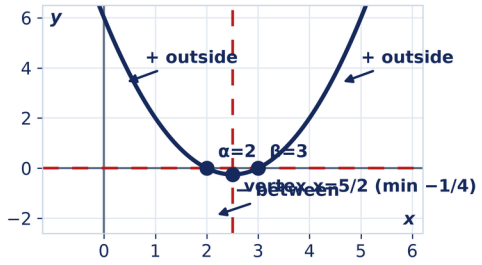
KEY RESULT

Sign analysis — the picture that solves every inequality

With real roots $\alpha < \beta$: for $a > 0$ the expression is **positive outside** $[\alpha, \beta]$ and **negative between** them ("outside positive, inside negative" for an upward parabola). For $a < 0$ all signs reverse. If $D < 0$ the sign never changes: it is the sign of a everywhere. Solve $ax^2 + bx + c > 0$ by roots-then-intervals, never by testing a random middle point. $x^2 - 5x + 6 < 0 \Leftrightarrow 2 < x < 3$.

y = x²−5x+6 (a>0)

Sign of a quadratic: + outside the roots, − between (a>0)



KEY RESULT

Never-negative / always-positive conditions

$ax^2+bx+c \geq 0$ for **all** real $x \Leftrightarrow a>0$ **and** $D \leq 0$. Similarly ≤ 0 for all $x \Leftrightarrow a<0$ and $D \leq 0$. Both conditions are needed: $D \leq 0$ with $a<0$ gives an always-negative expression. This pair is the standard engine for parameter ranges and later powers the discriminant method for ranges of rational functions.

Range on all of $\mathbb{R}$: $a>0$: $\left[\frac{4ac-b^2}{4a}, \infty\right)$; $a<0$: $\left(-\infty, \frac{4ac-b^2}{4a}\right]$. On a restricted interval, check whether the vertex lies inside; if not, the extremum is at an endpoint.

COMMON TRAP

Common traps — the expression

- Calling $\frac{4ac-b^2}{4a}$ a maximum when $a>0$ — it is a minimum; the sign of a decides.
- Sign pattern quoted without the sign of a.
- "Always positive" written with $D<0$ but strict/non-strict mixed up: >0 for all x needs $a>0, D<0$; ≥ 0 allows $D=0$ (touching zero at one point).
- On interval range problems, using the vertex value when the vertex is outside the interval.

INTUITION

AM–GM is this sheet in disguise

For $x>0$, $x+\frac{k}{x} \geq 2\sqrt{k}$ is $(\sqrt{x}-\frac{\sqrt{k}}{\sqrt{x}})^2 \geq 0$ — a perfect square, i.e. a $D=0$ quadratic. Least value of $x+\frac{4}{x}$ is 4 at $x=2$. Most "quadratic-flavoured" extrema are a vertex or a square in hiding.

LESSON 04 / 07

The three ingredients

Let $f(x)=ax^2+bx+c$, $a \neq 0$, roots $\alpha \leq \beta$ real. Every location condition is assembled from three ingredients only:

KEY RESULT

The location toolkit

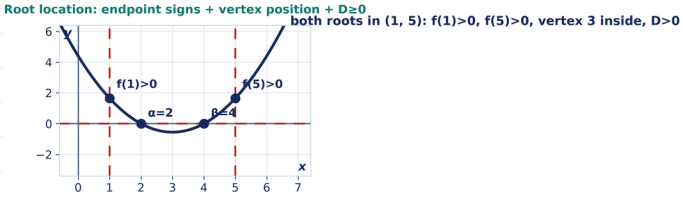
1. **Reality:** $D \geq 0$.
2. **Vertex position:** $x_v=-\frac{b}{2a}$ relative to the interval (the axis must sit where the roots are claimed to be).
3. **Endpoint sign:** $a \cdot f(k)$ at endpoints. The product $af(k)>0$ means k lies outside $[\alpha, \beta]$ (same sign region as far away); $af(k)<0$ means k lies strictly between the roots; $f(k)=0$ means k **is** a root.

Multiply by a to kill case-splitting on the sign of a; always work with $af(k)$.

KEY RESULT

The standard configurations (learn as pictures, not formulas)

- **Both roots $>k$:** $D \geq 0$, $af(k)>0$, $x_v>k$.
- **Both roots $<k$:** $D \geq 0$, $af(k)>0$, $x_v<k$.
- **k between the roots** ($\alpha<k<\beta$): $af(k)<0$ alone (this forces $D>0$ automatically).
- **Both roots in (k_1, k_2):** $D \geq 0$, $af(k_1)>0$, $af(k_2)>0$, $k_1<x_v<k_2$.
- **Exactly one root in (k_1, k_2):** $f(k_1)f(k_2)<0$ (then $D>0$ is automatic), plus separately check endpoint roots $f(k_i)=0$.
- **Roots of opposite signs:** $af(0)=ac<0$ alone (i.e. $P=\frac{c}{a}<0$); again $D>0$ is automatic.
- **Both roots positive:** $D \geq 0$, $S>0$, $P>0$. Both negative: $D \geq 0$, $S<0$, $P>0$. (This is the $k=0$ case of the first two bullets — Vieta form.)



WORKED EXAMPLE

Worked — both roots exceed 1

Both roots of $x^2 - 2kx + k^2 - 1 = 0$ exceed 1. $D = 4 > 0$ always; the roots are $k \pm 1$, so the smaller exceeds 1 iff $k - 1 > 1$, i.e. $k > 2$. Toolkit cross-check: $x_v = k > 1$ and $af(1) = k^2 - 2k = k(k-2) > 0$ give $k > 2$ or $k < 0$; intersecting with $k > 1$ leaves $k > 2$ ✓. Solving directly was cleaner here only because the roots are explicit — use the toolkit when they are not.

WORKED EXAMPLE

Worked — exactly one root in (0, 2)

$f(x) = x^2 - 3x + k$ has exactly one root in $(0, 2)$; $f(0)f(2) = k(k-2) < 0$ gives $0 < k < 2$; endpoints must be checked individually: $k = 0$ gives roots $0, 3$ — none inside, excluded; $k = 2$ gives roots $1, 2$ — the root 1 **is** inside, so $k = 2$ is included. Answer $0 < k \le 2$, with the endpoint discussion stated — losing it (in particular, mechanically excluding both endpoints) is the standard half-mark.

COMMON TRAP

Common traps — location

- Forgetting any one of the three ingredients: $D \ge 0$ and $af(k) > 0$ without the vertex condition also admits both roots on the other side of k.
- Using $f(k) > 0$ instead of $af(k) > 0$ when a may be negative or contains a parameter.
- Adding $D > 0$ redundantly (and then mishandling equality) when $af(k) < 0$ already forces distinct real roots.
- Strict vs non-strict: "in (k_1, k_2)" excludes endpoint roots; "in $[k_1, k_2]$" requires checking $f(k_i) = 0$ cases separately.
- When the leading coefficient contains the parameter, the degenerate linear case must be tested on its own.

ASSUMPTIONS & CONVENTIONS

Assumptions behind the toolkit

Real coefficients and real roots throughout; "root in interval" claims are meaningless for non-real roots. If a depends on the parameter, first confirm the equation stays quadratic on the claimed parameter range.

LESSON 05 / 07

The standard substitutions

KEY RESULT

Recognise the shape, name the substitution

- $ax^4 + bx^2 + c = 0$ (biquadratic): $t = x^2$, keep only $t \ge 0$.
- $a\left(x^2 + \frac{1}{x^2}\right) + b\left(x + \frac{1}{x}\right) + c = 0$: $t = x + \frac{1}{x}$, using $x^2 + \frac{1}{x^2} = t^2 - 2$. Note $|t| \ge 2$ for real $x \ne 0$ — reject $|t| < 2$.
- $a^{2x} + b\,a^x + c = 0$ (and $\sin$, $\log$ analogues): $t = a^x > 0$ — require $t > 0$.
- $(x+a)(x+b)(x+c)(x+d) = k$ with $a + d = b + c$: pair outer with outer so both products share the same $x^2 + (\text{common})x$ part; set t to that shared part.
- Radical equations: isolate the radical, square, solve — then **verify every candidate in the original equation**.
- Anything of the form $g(x)^2 + b\,g(x) + c = 0$: $t = g(x)$, with the range of g as the admissibility filter.

WORKED EXAMPLE

Worked — the four canonical types

Biquadratic: $x^4 - 5x^2 + 4 = 0$: $t = x^2$ gives $t = 1, 4$, so $x = \pm 1, \pm 2$.

Exponential: $2^{2x} - 5 \cdot 2^x + 4 = 0$: $t = 2^x > 0$ gives $t = 1, 4$ (both admissible), so $x = 0, 2$.

Reciprocal: $x^2 + \frac{1}{x^2} = 7$: with $t = x + \frac{1}{x}$, $t^2 - 2 = 7$, $t = \pm 3$; then $x = \frac{3 \pm \sqrt{5}}{2}$ or $x = \frac{-3 \pm \sqrt{5}}{2}$.

Radical: $\sqrt{x+3} = x - 3$: squaring gives $x^2 - 7x + 6 = 0$, candidates $x = 1, 6$. Check: $x = 1$ gives $2 = -2$ ✗ (also the RHS is negative while a square root never is — rejectable *before* squaring); $x = 6$ gives $3 = 3$ ✓. Answer: $x = 6$ only.

COMMON TRAP

Common traps — reducible equations

- Squaring is not reversible: extraneous roots are the rule, not the exception. Verification is part of the method.
- Dividing by an expression that can vanish loses roots: $(x+1)^2 = 4(x+1)$ divided by $x + 1$ loses $x = -1$; the substitution $t = x + 1$ gives $t = 0, 4$, i.e. $x = -1, 3$.
- Admissibility forgotten: $t = x^2 < 0$ or $t = 2^x \le 0$ or $|x + \frac{1}{x}| < 2$ candidates must be discarded even though the quadratic "gave" them.
- For $\sqrt{A} = B$, record $B \ge 0$ and $A \ge 0$ first — often half the candidates die instantly.

STANDARD APPLICATION

Why quadratics are the reduction target

Biquadratics, radical, exponential, logarithmic and many trigonometric equations all collapse to a quadratic after one well-chosen substitution. The exam skill is not solving the quadratic — it is (1) spotting the repeated block $g(x)$, (2) writing its admissible range, (3) filtering. Marks are lost at steps 2–3, almost never at step 1.

LESSON 06 / 07

A common root

KEY RESULT

Subtract first, condition later

If $ax^2 + bx + c = 0$ and $a'x^2 + b'x + c' = 0$ share a root α, subtracting kills the x^2 term (after scaling to equalise leading coefficients): the common root satisfies a **linear** equation, and is usually immediate. The textbook condition

$$(bc' - b'c)^2 = (ab' - a'b)(ac' - a'c)$$

is the same statement after elimination — correct but slow; reach for it only when subtraction degenerates (identical or proportional equations, which share *both* roots: coefficients proportional, $\frac{a}{a'} = \frac{b}{b'} = \frac{c}{c'}$).

WORKED EXAMPLE

Worked — common roots

$x^2 - 5x + 6 = 0$ and $x^2 - 4x + 3 = 0$: subtract, $-x + 3 = 0$, so the common root is $x = 3$ (check: $9 - 15 + 6 = 0$ ✓, $9 - 12 + 3 = 0$ ✓).

If $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ ($a \ne b$) share a root: subtracting gives $(a - b)(x - 1) = 0$, so the common root is 1 and substituting gives $a + b + 1 = 0$.

Transformed roots

KEY RESULT

Build from new S and new P — or substitute and skip the algebra

- Roots $\alpha+k, \beta+k$: new sum $S+2k$, new product $P+kS+k^2$. Equivalent shortcut: replace x by $x-k$ in the original equation.
- Roots $\frac{1}{\alpha}, \frac{1}{\beta}$ $(P \neq 0)$: new sum $\frac{S}{P}$, new product $\frac{1}{P}$. Shortcut: reverse the coefficients — $cx^2+bx+a=0$.
- Roots $k\alpha, k\beta$: sum kS, product k^2P. Shortcut: replace x by $\frac{x}{k}$ and clear.
- Roots α^2, β^2: sum S^2-2P, product P^2.
- Roots α^3, β^3: sum S^3-3PS, product P^3.

WORKED EXAMPLE

Worked — transformations

From $x^2-5x+6=0$ $(S=5,\ P=6)$: roots $\alpha+1, \beta+1$ give sum 7, product $6+5+1=12$, equation $x^2-7x+12=0$ (actual roots $3, 4$ ✓). Reciprocals give sum $\frac{5}{6}$, product $\frac{1}{6}$, equation $6x^2-5x+1=0$ (actual $\frac{1}{2}, \frac{1}{3}$ ✓).

COMMON TRAP

Common traps — common and transformed roots

- The squared-roots equation has product P^2, not P.
- Reciprocal transformation silently assumes neither root is 0; if $c=0$ the "reciprocal equation" drops degree.
- Common-root condition memorised with the factors in the wrong pairing — derive by elimination if unsure, or just subtract.
- Two quadratics can share both roots without being identical: proportional coefficients are enough.

STANDARD APPLICATION

The graph view (what the algebra is saying)

$y=ax^2+bx+c$: vertex $\left(-\frac{b}{2a}, \frac{4ac-b^2}{4a}\right)$, axis $x=-\frac{b}{2a}$, meets the axis iff $D \geq 0$, opens up iff $a>0$. Shifting roots by k translates the graph; reversing coefficients reflects the root set by $r \mapsto \frac{1}{r}$. Every transformation above is a picture before it is a formula.

LESSON 07 / 07

The five shapes every parameter problem takes

KEY RESULT

Shape recognition

1. **Nature prescribed** ("real/distinct/equal/non-real roots"): write D and solve the inequality — plus the $a \neq 0$ degeneracy check.
2. **Condition between roots** ("one root is double the other", "roots differ by 2", "reciprocals of each other"): Vieta plus the extra relation. Reciprocals condition: $P=1$, i.e. $c=a$. One root double: roots $t, 2t$, so $S=3t$, $P=2t^2$, eliminate: $9P=2S^2$ (for the monic-scaled S, P).
3. **Sign prescribed** ("both positive", "opposite signs"): sheet 04, Vieta form: $D \geq 0$, signs of S and P.
4. **Location prescribed:** sheet 04 toolkit, three ingredients.
5. **Range/extremum asked:** vertex (sheet 03), or write $y=\frac{\text{quadratic}}{\text{quadratic}}$, cross-multiply to a quadratic in x and demand $D \geq 0$ — the **discriminant method** for ranges.

WORKED EXAMPLE

Worked — discriminant method for a range

Range of $y = \frac{x^2-3x+4}{x^2+3x+4}$ ($x \in \mathbb{R}$; denominator > 0 always since its $D = 9 - 16 < 0$).

Cross-multiply: $(y-1)x^2 + 3(y+1)x + 4(y-1) = 0$. Real x exists iff $D = 9(y+1)^2 - 16(y-1)^2 \geq 0$, i.e. $-7y^2 + 50y - 7 \geq 0$, i.e. $(7y-1)(y-7) \leq 0$, so $\frac{1}{7} \leq y \leq 7$. (The case $y = 1$ separately gives $x = 0$, consistent.) Equality endpoints are attained, so the range is the closed interval.

KEY RESULT

Recap — the chapter on one card

Formula $x = \frac{-b \pm \sqrt{D}}{2a}$; D cases; $S = -\frac{b}{a}$, $P = \frac{c}{a}$; $\alpha^2 + \beta^2 = S^2 - 2P$; $\alpha^3 + \beta^3 = S^3 - 3PS$; $(\alpha - \beta)^2 = S^2 - 4P = \frac{D}{a^2}$; vertex $\left(-\frac{b}{2a}, \frac{4ac-b^2}{4a}\right)$; sign: a's sign outside, opposite between; always ≥ 0 iff $a > 0, D \leq 0$; location = D + $af(k)$ + vertex; reducible = substitute, filter by range, verify; common root = subtract; transformed = new S, new P.

COMMON TRAP

The ten mistakes that cost the marks (chapter checklist)

1. $D = b^2 + 4ac$ sign slip. 2. "$D < 0 \Rightarrow$ no roots." 3. $(\alpha - \beta)^2 = S^2 - 2P$. 4. Vertex value called max when $a > 0$. 5. Sign pattern ignoring sign of a. 6. Location: dropping one of D, $af(k)$, vertex. 7. Parameter in a: degenerate linear case unchecked. 8. Squared-roots product written P not P^2. 9. No verification after squaring. 10. Dividing away a vanishing factor and losing a root.

INTUITION

Forward map — where quadratics reappear

Vieta is the seed of symmetric polynomials (Part II here, and Newton's identities); the vertex is the seed of optimisation and of the discriminant range method used again in conic ranges; roots-of-unity filters, quadratic residues and Vieta jumping all begin with "read this symmetric condition as a quadratic in one variable".